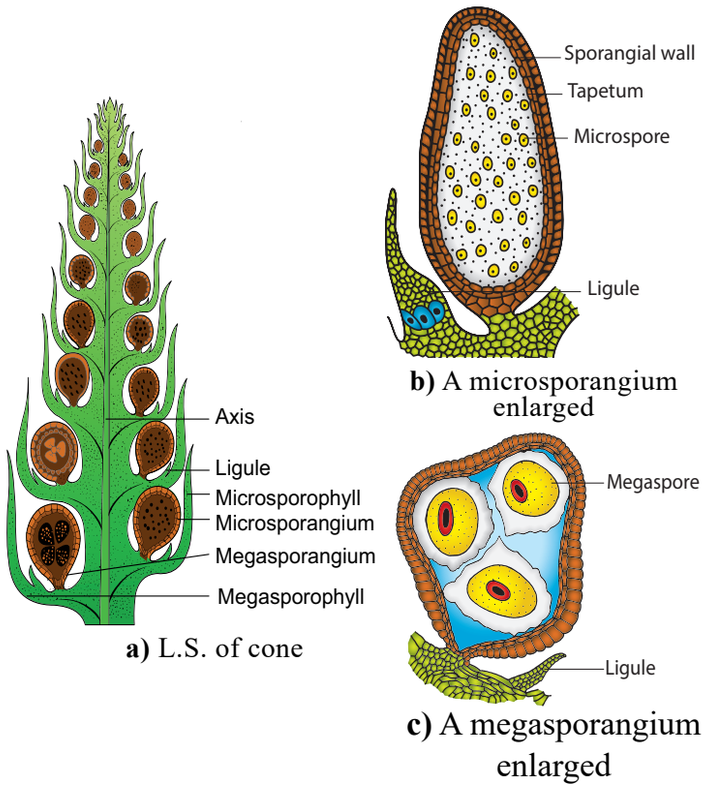



**a)** L.S. of cone

**b)** A microsporangium enlarged

**c)** A megasporangium enlarged

**Figure 2.19:** Reproduction in *Selaginella*

### Sexual reproduction

During sexual reproduction spores are produced (Figure 2.19). *Selaginella* is heterosporous and produces two types of spores namely microspores in microsporangium and megaspores in megasporangium. The sporangia are borne singly in the axils of microsporophyll and megasporophyll respectively. The sporophylls are arranged spirally around a central axis and aggregate to form strobili or cones. Variations in the distribution of microsporangia and megasporangia among the species are seen. In S. *selaginoides* and *S. rupestris* megasporangia are present in the basal part of the cone. *S. kraussiana* possesses a single megasporangium at the base of the strobilus. In *S. inaequifolia* one side of the strobilus bear only megasporangia and other microsporangia. Separate strobili for microsporangia and megasporangia are present in *S. gracilis*. and *S. atro-viridis*.

The development of sporangium is of eusporangiate type. The sporangial initial divides periclinally to form outer jacket initials and inner archesporial initials. The archesporial initials by repeated anticlinal and periclinal divisions form sporogenous cells. Microspore mother cells of microsporangium undergo reduction division to produce halpoid microspores. Similarly the megaspore mother cell undergoes reduction division to produce 4 haploid megaspores. The microspore and megaspore represent the male and female gametophyte and germinate inside the sporangium. The microspores produce biflagellate antherozoids. Archegonia develop in the female gametophyte. The antherozoids swim in water and reach the archegonium. Fertilization brings the fusion of male and female nucleus which result in the formation of a diploid zygote. The diploid zygote represents the first cell of sporophyte. It undergoes several mitotic division to form embryo. The embryo develops into a mature sporophytic plant.

In the life cycle of *Selaginella* alternation of sporophytic and gametophytic generation is present (Figure 2.20).

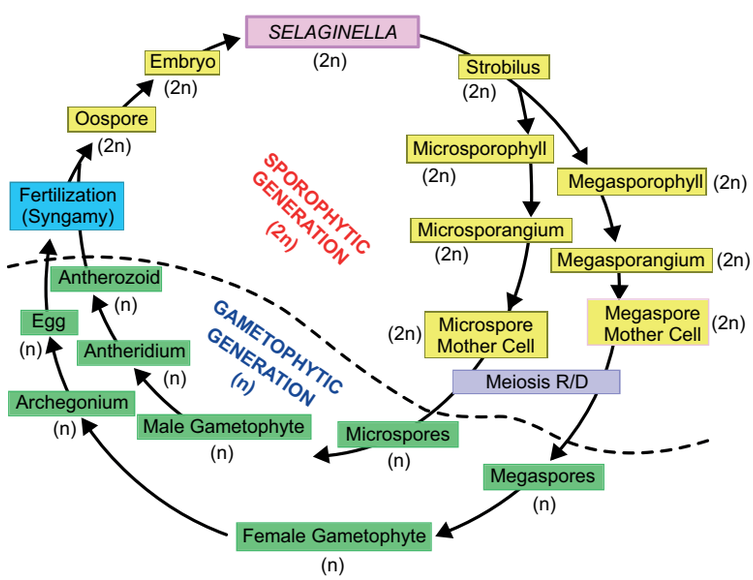



**Figure 2.20:** Life cycle of *Selaginella*



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